Ziwei Liu

DATA SCIENCE UNDERGRADUATE · MACHINE LEARNING RESEARCH

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PROFILE

Year 2 Data Science and Technology undergraduate at HKUST with hands-on research experience in reward model training, world model distillation, and VLA/LLM training. Current focus on mechanistic interpretability of large models.

EDUCATION

The Hong Kong University of Science and Technology — Year 2 · Hong Kong

BSc in Data Science and Technology

CUMULATIVE GPA **3.92** / 4.30

HONOURS

- **HKUST Continuing Scholarship** — awarded for sustained academic excellence
- **S. S. Chern Class** — member of HKUST's selective honours class

PUBLICATIONS

How Much Opaque Serial Depth Does a Loop Buy? A Preregistered Study of No-CoT Reasoning in Small Looped Transformers — 2026

Ziwei Liu · *Preprint*, October 2026 — PDF

Reaction Without Attribution: Hijacking a VLA's Actions Reveals No Linearly Readable Efference Copy — 2026

Ziwei Liu · *NeurIPS 2026 RoboPAD Workshop* (accepted) — OpenReview · PDF

RESEARCH

Undergraduate Research Assistant (UGRA) — Jan 2026 – Aug 2026

PeiLab, HKUST

- Trained reward models and ran model-training experiments in PyTorch.
- Distilled the DreamDojo world model, transferring learned capabilities into a compact student model.
- Built and maintained end-to-end training pipelines for VLA and LLM-based models.

SKILLS

PROGRAMMING	Python
FRAMEWORKS	PyTorch · TransformerLens
MACHINE LEARNING	Reward model training · World model distillation · VLA and LLM training
MECH INTERPRETABILITY	Sparse autoencoders and dictionary learning · Activation patching and causal tracing · Logit lens and attribution analysis · Circuit discovery · Linear probing of internal representations